

A Ten-Paper Fixed-Shift Route Audit: Dependency Closure, Endpoint Losses, and the Primitive-Plane Frontier

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

We audit the ten-paper block TPC-63 through TPC-72, using the completed results of TPC-63 through TPC-71 as input. The purpose is not to promote interface theorems into arithmetic conclusions, but to determine exactly which dependencies are closed, which proof mechanisms have failed, and which literal estimates remain necessary before a fixed-shift endpoint can be claimed.

The represented branch now has an exact chain from canonical cofactor provenance through terminal activation, activated-range Grams, actual pair-codegree geometry, and componentwise determinant mass. TPC-68 gives exact abstract certificates for the complete signed missing-native operator. TPC-69–TPC-71 then isolate one mixed scalar sector and reduce its anchored four-corner contributions through erasure-boundary words, primitive cofactor coordinates, and a divisor-lifted two-dimensional Abel certificate. In particular, the first cross-key sign becomes $\mu(a)\mu(b)\mu(u)\mu(v)$, is constant along each affine prime ray, and leaves a primitive-plane kernel. The identity

$$\mathbf{1}_{(u,v)=1}\mu(u)\mu(v) = \mu(uv)$$

and a diagonal divisor lift reduce regular kernels to restricted Mertens sums. Fixed-divisor prime number theory then gives cancellation when every fixed- d mixed variation is $O_d(\|K\|_\infty)$. It does not give the mass-normalized variation or divisor tail required by the actual physical kernel.

The audit also closes a tempting but type-incorrect physical promotion of this certificate. The TPC-70 primitive-plane target is an anchored four-corner scalar contributing to $\text{tr}(ER)$, not a two-corner matrix entry of the erasure operator E . Consequently, even a sharp bound for every audited primitive-plane target does not by itself imply an entrywise majorant, a row-star estimate, a Perron bound, or an operator norm estimate. The literal anchor multiplicity must be retained, and the four-corner normalization must not be applied a second time.

We prove a dependency theorem for the audited chain, reconcile its exponent notations into one no-double-charge ledger, and state exact kill criteria. Algebraic identities and isometric or subpower-conditioned relabelings have zero fixed-power cost. Every non-isometric physical comparison exposed by them must be charged exactly once in a total strictly below $1/400$. The missing-native perturbative gate $\sigma_M > \nu_M + \lambda_{\text{rep}}$ and the nuisance condition $\|C_N\| = o(1)$ remain separate. As of TPC-71 there is no completed L2 estimate satisfying the full fixed-shift quantifier. The next stage may estimate the physical primitive-plane variation and tail to control the first mixed trace, but an operator theorem additionally needs one of three new bridges: a direct two-corner representation, row-local anchored bounds dominating the relevant stars, or uniformly controlled complete decorated moments. Terminal activation and the represented lower frame must advance in parallel. No fixed-2, parity-breaking, prime-pair, or twin-prime conclusion is drawn.

Keywords: fixed shift; proof audit; native Gram; terminal activation; determinant mass; Möbius function; two-dimensional Abel summation; endpoint ledger.

Convention. Every physical statement retains one fixed nonzero shift h_0 , literal coefficients, actual active support, complete native labels, and raw atomic normalization. No result is specialized to $h_0 = 2$.

MSC 2020: Primary 11N37, 11A25; Secondary 47A30, 15A18.

1 Introduction

A long proof program can accumulate exact lemmas while losing track of the quantifier needed at its endpoint. This danger is particularly acute in a fixed-shift parity problem. A theorem for an abstract sign matrix, a fixed row pair, an averaged shift, or a smoothed coefficient family can be correct without estimating the literal growing operator. Likewise, an exact coordinate identity can remove a modeling ambiguity without producing any arithmetic saving.

TPC-63 through TPC-71 were designed to separate these roles [3–11]. They form two chains. The first follows the represented packet:

provenance $\longrightarrow$ activation $\longrightarrow$ represented Gram $\longrightarrow$ collision geometry $\longrightarrow$ determinant mass.

The second begins with the complete missing-native packet and then follows one mixed scalar sector:

signed native operator $\longrightarrow$ first mixed trace $\longrightarrow$ primitive cofactor plane
 $\longrightarrow$ two-dimensional variation.

The represented branch and the complete missing-native operator meet only at physical reassembly. The descendant primitive-plane chain does not itself return from a scalar trace sector to the missing-native operator. A result on one branch does not silently supply a missing hypothesis on the other.

1.1 Why the present audit is necessary

Three developments make a route audit timely.

First, TPC-63 proves that the inherited single-band freezing certificate cannot reach the critical high face within the endpoint budget [3]. The failure concerns a proof mechanism, not the actual weighted mass. Second, TPC-66 gives a genuine fixed-shift arithmetic support theorem: the reduced codegree of every fixed row pair is subpower on the critical block [6]. Yet TPC-67 shows why pair sparsity is not a represented lower-frame theorem: long cores and phase holonomy remain [7]. Third, TPC-68–TPC-71 progressively remove false sources of Möbius randomness. Same-residual signs freeze, one-key mixed traces vanish, the complete quadratic erasure response is nonnegative, and the surviving primitive sign is constant along each prime ray [8–10]. TPC-71 nevertheless opens a new positive door: regular primitive-plane kernels cancel by an exact divisor lift and two-dimensional Abel summation [11].

There is also a necessary type boundary. The primitive-plane object inherited from TPC-70 contains one erasure edge and one same-residual anchor. It is therefore a four-corner contribution to $\text{tr}(ER)$, whereas an entry of E is a two-corner sum. Treating the former as an entrywise majorant for the latter changes the mathematical object, loses the anchor index, and can apply the row normalization twice. TPC-71 correctly retains the anchored trace target. The present audit promotes that boundary to an explicit non-substitution law: the direct primitive-plane-to-star/Perron promotion is unavailable.

These are real advances, but of different logical types. The present paper classifies them before a local improvement is mistaken for an endpoint theorem.

1.2 Main conclusions

The audit has seven conclusions.

- (1) TPC-63–TPC-71 close a sequence of necessary L0/L1 interfaces. No paper in the block proves a complete L2 cancellation estimate.
- (2) The represented and missing-native branches are logically independent until shorted reassembly. Determinant mass does not bound the missing-native defect, and primitive-plane cancellation does not prove a represented lower frame.
- (3) Several routes are permanently closed: the inherited single-band support certificate, same-residual Möbius randomness, the one-key mixed trace, and a negative quadratic erasure correction.
- (4) TPC-71 proves cancellation relative to ambient area for regular kernels. The physical denominator is actual atomic mass, so the mass-normalized mixed variation and diagonal tail are new hypotheses, not corollaries.

- (5) The primitive-plane target is an anchored contribution to $\text{tr}(ER)$, not an entry of E . A separate direct two-corner representation, row-local anchored-star bridge, or complete-moment argument is required before an operator theorem can be invoked.
- (6) Algebraic identities and isometric or subpower-conditioned relabelings cost zero fixed power. Every non-isometric physical comparison must be charged once in a common ledger strictly below $1/400$.
- (7) The next routes have explicit success and failure tests. One route controls the literal anchored primitive-plane kernel strongly enough to bound the complete first mixed trace. The other constructs a genuine operator-level carrier through direct entries, row-local anchored stars, or complete moments. A global-trace estimate alone is insufficient.

1.3 Organization

Section 2 defines the audit levels and physical invariants. Sections 3 to 5 audit TPC-63–TPC-69. Section 6 records the TPC-70–TPC-71 frontier. Section 7 proves the dependency theorem. Section 8 reconciles exponent bookkeeping. Sections 9 and 10 separate closed doors from local certificate failures. The revised route and final claim ledger appear in sections 11 and 12.

2 Audit levels and invariant physical data

2.1 The three levels

Definition 2.1 (Audit levels). We use the following labels.

- L0 A finite-dimensional, combinatorial, or abstract analytic theorem valid for the declared object, before specialization to the literal growing physical family.
- L1 An exact identification or unconditional support statement on the literal fixed- h_0 carrier, retaining its coefficients, labels, support, and normalization, but not proving the required signed asymptotic saving.
- L2 A uniform arithmetic estimate for the actual physical coefficients in the fixed-shift growing family, with sufficient strength and quantifiers to enter the endpoint chain.

An L1 upper count can use number theory and still remain L1 when it controls support geometry rather than the required signed operator. Conversely, a theorem need not solve the final problem to be L2; a fixed-power saving on one literal endpoint arrow is already an arithmetic advance if its scope matches the physical chain.

2.2 The six invariant checks

Every proposed promotion to L2 is tested against:

- (A1) the literal physical coefficients;
- (A2) one prescribed fixed nonzero shift h_0 ;
- (A3) raw physical atomic normalization;
- (A4) a canonical or explicitly justified representation;
- (A5) the actual active support, with no coefficient-dependent deletion;
- (A6) the complete endpoint loss, strictly below $1/400$.

Dropping any check changes either the object, the norm, or the quantifier.

2.3 Three kinds of progress

Definition 2.2 (Counted progress). A result counts as forward proof progress in this audit if it does at least one of the following:

- (i) improves a concrete exponent for literal physical coefficients;
- (ii) closes a necessary interface in the dependency chain; or
- (iii) rules out a route by an exact physical obstruction or a sharp non-implication theorem.

The definition recognizes structural progress without relabeling it as arithmetic cancellation. Numerical stability, generic phases, randomized signs, or finite fixed-dimensional experiments do not satisfy [definition 2.2](#) unless they yield an exact certificate.

2.4 Uniform and distinguished scopes

Several papers admit two honest coefficient scopes.

- A *uniform* theorem holds for every vector in the declared coefficient class and therefore requires rowwise or operator control.
- A *distinguished* theorem holds for one fixed physical vector chosen independently of the later estimate.

The second cannot be entered as the first. Likewise, a fixed-pair estimate, a density-one family, or an average over shifts cannot be substituted for the complete fixed- h_0 growing-row statement.

Proposition 2.3 (Audit monotonicity). *An exact L0 identity plus an L1 crosswalk does not imply an L2 estimate without a quantitative arithmetic input on the crosswalked object.*

Proof. Take any finite incidence or matrix identity in the audited chain and fill the same support with two coefficient systems having different phases. TPC-65–TPC-69 contain explicit cycle, row-capacity, star, and holonomy countermodels showing that the same structural data can yield incompatible spectral or cancellation behavior [5–9]. Therefore the structural identity and physical dictionary do not determine the missing quantitative estimate. $\square$

3 TPC-63–TPC-64: provenance and true activation

3.1 TPC-63: the carrier cannot be repaired by declaration

TPC-63 distinguishes a literal full incidence from its pair projection [3]. Three faithfulness questions are separate:

- (i) preservation of the pre-terminal raw form;
- (ii) preservation of terminal atoms; and
- (iii) preservation of the grouped physical operator.

An active spurious cross can pass one test and fail another. When a terminal-faithful pair completion exists, there is a unique least relation; when it does not, the literal incidence hypergraph is the canonical faithful object.

Let $W_H(m)$ be native-high parent mass and $W_A(m)$ the part aligned with the terminal predicate. Inside the aligned mass, let the declared retained relation supply completion ratio $c_{\text{comp}}(m)$. TPC-63 proves the exact factorization

$$\frac{W_{\text{exposed}}(m)}{W_H(m)} = \frac{W_A(m)}{W_H(m)} c_{\text{comp}}(m) \quad (1)$$

whenever $W_H(m) > 0$. Passing to the maximal eligible relation may make $c_{\text{comp}}(m) = 1$, but it cannot improve the alignment factor.

The same paper proves a negative L1 result for the inherited single-band mechanism. The balanced band certified by the old freezing inequality is eventually disjoint from the critical high face. Retuning that sufficient inequality to the critical cofactor scale already costs more than the entire 1/400 allowance. Even after an independently justified retuning, a completely omitted dyadic block makes the inherited support-cardinality/Brun–Titchmarsh certificate yield no positive fixed-power saving.

Proposition 3.1 (Scope of the TPC–63 no-go). *The TPC–63 negative result closes the named single-band support certificate. It does not imply that the actual omitted weighted mass is large.*

Proof. The certificate replaces literal weights by support and harmonic majorants. Its failure shows only that those upper data cannot establish the desired lower ratio. A coefficient-sensitive weighted distribution estimate or exact signed shorting uses information absent from that certificate and is not contradicted. $\square$

Orthogonally retaining all nonempty dyadic labels gives exact raw-energy coverage at $O(\log X)$ label complexity. The later erasure of these labels is a physical map, not a harmless relabeling.

3.2 TPC–64: activation, inflation, and capacity

TPC–64 defines, before the free row coefficient is chosen, the parent mass P_m , terminal-support mass H_m , and shaped terminal energy T_m [4]. On a row where both denominators are nonzero,

$$\frac{T_m}{P_m} = \frac{H_m}{P_m} \frac{T_m}{H_m}. \quad (2)$$

The two factors are terminal-support alignment and terminal multiplier intensity.

A parent-complete pair lift can add terminal-null coordinates. Its repaired parent is

$$\tilde{P}_m = P_m + \Delta_m, \quad T_m \text{ unchanged}. \quad (3)$$

Thus a pair completion may preserve the numerator while diluting its denominator by an arbitrary polynomial factor. A newly created parent row with $P_m = 0 < \tilde{P}_m$ makes uniform parent domination infinite. The literal-active-hypergraph baseline avoids this inflation exactly.

For $g_m = T_m/\tilde{P}_m$, let δ_η be the repaired-parent fraction on rows $g_m \geq \eta$, and let κ_X measure the occupied coefficient capacity. TPC–64 proves the sharp lower certificate

$$\frac{T(z)}{\tilde{P}(z)} \geq \eta \left(1 - \frac{1 - \delta_\eta}{\kappa_X} \right)_+. \quad (4)$$

The right side is positive exactly when $\kappa_X > 1 - \delta_\eta$.

3.3 The first unresolved arithmetic arrow

There are two honest next inputs:

$$T_m \geq X^{-\lambda_{\text{row}} + o(1)} \tilde{P}_m \quad \text{for every active row}, \quad (5)$$

$$\eta_X \left(1 - \frac{1 - \delta_{\eta_X}}{\kappa_X(z_*)} \right)_+ \geq X^{-\lambda_{\text{term},*}^{\text{eff}} + o(1)} \quad \text{for one declared } z_*. \quad (6)$$

Neither is proved in TPC–63 or TPC–64. A row with $T_m = 0 < \tilde{P}_m$ kills (5) on every coefficient class containing that row, but it does not kill a separately justified distinguished scope.

Remark 3.2 (Audit classification). The incidence factorizations, minimax constants, inflation laws, and capacity extremizers are L0. Their literal fixed- h_0 specialization is L1. Establishing either (5) or (6) at endpoint strength would be L2.

4 TPC–65–TPC–67: represented range, support, and determinants

4.1 TPC–65: the actual activated range

Let L_X be the terminal map and

$$Y_X = \text{Ran } L_X.$$

TPC–65 places the represented comparison on Y_X , rather than on a free ambient atom space [5]. If $A_X : Y_X \rightarrow H_X$ is the represented map, its exact lower constant is

$$\beta_X = \inf_{0 \neq y \in Y_X} \frac{\|A_X y\|^2}{\|y\|^2} = \lambda_{\min}(K_X), \quad (7)$$

where K_X is the normalized activated row Gram with literal residual-fiber entries.

Terminal-faithful canonical repair leaves downstream Grams unchanged when both descriptions factor through the same terminal map. Parent inflation is therefore charged in the parent-to-terminal arrow, not again in β_X .

For orthogonally labeled blocks A_ν ,

$$\text{Ker} \left(\bigoplus_{\nu} A_\nu \right) = \bigcap_{\nu} \text{Ker } A_\nu. \quad (8)$$

Singular blocks can fuse into a positive global Gram. Coherently erasing their labels instead creates cross-block terms and requires a separate fusion-angle theorem. TPC–65 also gives a sharp counterexample to transporting physical row capacity into spectral capacity.

On one common carrier, define activation, represented, and reassembly constants $\tau_X, \beta_X, \delta_X$. The exact compatible product is

$$\Gamma_X \geq \tau_X \beta_X \delta_X. \quad (9)$$

This is a compatibility law, not a proof that any physical factor is favorable.

4.2 TPC–66: actual pair geometry

TPC–66 opens the literal entries of K_X [6]. A fixed physical row contributes at most one terminal atom to a fixed residual output key. After row normalization,

$$(K_X)_{mm} = 1 \quad (10)$$

for every active row.

For distinct literal rows m_1, m_2 , factor exchange and simultaneous CRT give the reduced pair-codegree bound

$$c_{m_1, m_2}^{\text{red}}(R) \ll_{h_0} X^{o(1)} \left(1 + \frac{J}{R} \right). \quad (11)$$

Consequently,

$$c_{m_1, m_2}^{\text{red}}(R) = X^{o(1)} \quad (R \asymp J) \quad (12)$$

uniformly over fixed active row pairs.

This is a genuine arithmetic support theorem at one fixed h_0 , but its scope must be retained. It counts reduced arithmetic cells, not the raw γ -decorated intersection. A right fiber may still have polynomial degree. The exact bridge

$$|K_X(m, n)| \leq C_{m, n}^{\text{red}}(R) \sqrt{\rho_m(R) \rho_n(R)} \quad (13)$$

shows that a physical Gram estimate still needs row-energy diffuseness.

Private outputs, comparison matrices, weighted diagonal dominance, forest gauges, star spectra, and unique-matching minors give exact lower-frame certificates. A cycle with fixed support, degree, codegree, and entry magnitudes can nevertheless be singular or nonsingular according to its holonomy. Support sparsity alone is therefore not a lower-frame theorem.

4.3 TPC–67: determinant mass on actual components

TPC–67 factors every square actual minor by iterative degree-one peeling [7]. A residual core with no perfect matching has zero determinant. Relative to a declared core matching, all remaining terms are vertex-disjoint alternating-cycle families. Forest phases gauge away; cycle holonomies remain.

If $\mathcal{A}_\phi$ is the total absolute activity relative to matching ϕ , the phase-safe certificate has the form

$$|\det T| \geq P_{\text{for}} P_\phi (1 - \mathcal{A}_\phi)_+. \quad (14)$$

The threshold one is sharp for information consisting only of total activity.

For a synthesis matrix B with r unit columns and Gram $K = B^* B$, Cauchy–Binet yields

$$\mathfrak{D}(B) := \sum_{|I|=r} |\det B[I]|^2 = \det K, \quad \lambda_{\min}(K) \geq e^{-1} \mathfrak{D}(B). \quad (15)$$

The actual full-support collision components are orthogonal, so the useful bound is

$$\beta_X \geq e^{-1} \min_{C \in \mathcal{C}_X} \mathfrak{D}_C, \quad (16)$$

not a product over unrelated components.

Proposition 4.1 (Represented branch frontier). *The represented branch would close at fixed-power scale if one proved, uniformly over actual full-support components,*

$$\mathfrak{D}_C \geq X^{-\lambda_{\det} + o(1)}$$

with an exponent compatible with the endpoint ledger. TPC–66 does not imply this premise.

Proof. Equation (16) gives the implication. TPC–66 controls only reduced fixed-pair codegree. It supplies neither a full-output Hall matching, pivot lower bounds, long-cycle activity, nor γ -multiplicity. Hence its hypotheses do not determine $\mathfrak{D}_C$. $\square$

All determinant conversions are L0, and the full-output identification is L1. The displayed lower bound for the actual worst component is an open L2 target.

5 TPC–68–TPC–69: signed native operators and mixed words

5.1 TPC–68: the complete missing-native defect

TPC–68 retains every missing-active row and every still-orthogonal native label [8]. Physical atomic normalization gives

$$K_M = I + H, \quad H = H^*, \quad H_{mm} = 0. \quad (17)$$

The residual split is

$$H = R + E,$$

where R contains same-residual collisions and E contains unequal-residual erasure edges.

After a unitary row gauge, a literal off-diagonal entry exposes target signs

$$\mu(mj + h_0) \mu(nj + h_0).$$

On a same-residual collision this product reduces to a row gauge $\mu(d_m) \mu(d_n)$. Every pure R -closed walk therefore has target-sign product one. New signed information must use an E -edge or a literal complex phase.

TPC–68 proves exact pair, star, Perron, and even-walk certificates. If $\Delta_P \leq X^{\delta+o(1)}$ and each pair majorant is at most $X^{-\tau+o(1)}$, then

$$\|H\| \leq X^{\delta-\tau+o(1)}. \quad (18)$$

If instead the maximal ℓ^2 -star is $X^{-\sigma+o(1)}$, then

$$\|H\| \leq X^{\delta/2-\sigma+o(1)}. \quad (19)$$

For a growing even moment,

$$\text{tr } H^{2q_X} \leq X^{2q_X \nu + o(q_X \log X)} \implies \|H\| \leq X^{\nu+o(1)}. \quad (20)$$

The displayed premises are L2 targets and are not proved there.

5.2 TPC-69: the first exact cross-key holonomy

TPC-69 refines the split by complete native key:

$$R = \sum_i R_i, \quad E = \sum_i E_i. \quad (21)$$

Relative to the residual partition of key i , R_i is block diagonal and E_i is off block. Hence, for every polynomial P ,

$$\text{tr}(E_i P(R_i)) = 0. \quad (22)$$

The first mixed half-sector is therefore necessarily cross-key:

$$S_{2,1} = 2 \text{tr}(ER), \quad \text{tr}(ER) = \sum_{i \neq j} \text{tr}(E_i R_j). \quad (23)$$

Each term is a genuine four-corner rectangle, invariant under independent row and key unit gauges.

For

$$\text{tr}(R + tE)^k = \sum_{b=0}^k t^b S_{k,b},$$

the exact cyclic sector formula is

$$S_{k,b} = \frac{k}{b} \sum_{\substack{a_1 + \dots + a_b = k-b \\ a_\ell \geq 0}} \text{tr}(ER^{a_1} \dots ER^{a_b}). \quad (24)$$

On the squarefree physical carrier, source-row signs telescope around a closed decorated walk. The surviving target sign is the product of $\mu(s)$ over residual labels of odd erasure degree.

The complete two-erasure sector has a further obstruction. If λ_α are the eigenvalues of R , then

$$S_{2q,2} = q \sum_{\alpha, \beta} |E_{\alpha\beta}|^2 \frac{\lambda_\alpha^{2q-1} - \lambda_\beta^{2q-1}}{\lambda_\alpha - \lambda_\beta} \geq 0, \quad (25)$$

with the derivative value on the diagonal. The inequality follows because the odd power x^{2q-1} is increasing.

Corollary 5.1 (Signed-door reduction). *The following proposed sources of negative compensation are closed:*

- (i) *Möbius randomness internal to pure same-residual walks;*
- (ii) *one-key cancellation in the first mixed sector; and*
- (iii) *the complete quadratic erasure sector as a negative correction.*

The shortest remaining signed object is the cross-key rectangle in (23).

Proof. The first statement follows from row-gauge telescoping; the second from (22); the third from (25). $\square$

A small global $\text{tr}(ER)$ is not a row-star estimate. Even a complete linear-sector saving does not bound the full moment unless all remaining sectors are controlled at the correct scale.

6 TPC-70-TPC-71: the primitive-plane frontier

6.1 TPC-70: exact cofactor coordinates

Fix distinct physical rows m, n , one nonzero shift h_0 satisfying $(mn, h_0) = 1$, and one full same-residual anchor $\mathfrak{a}$. Put

$$H = h_0(n - m). \quad (26)$$

At a same-residual anchor, TPC-70 writes

$$d_m = ga, \quad d_n = gb, \quad (a, b) = 1. \quad (27)$$

Equality of residuals forces the complementary cofactors to have the form

$$R_m = bt, \quad R_n = at, \quad t \mid |H|. \quad (28)$$

For fixed data, the anchor times lie on an affine ray

$$j_R = j_{R,0} + abt z. \quad (29)$$

At an erasure edge, write

$$S_m = qu, \quad S_n = qv, \quad (u, v) = 1. \quad (30)$$

The determinant relation forces $q \mid |H|$, and admissible erasure times lie on

$$j_E = j_{E,0} + quv z. \quad (31)$$

The erased residuals are unequal exactly when

$$(u, v) \neq (b, a). \quad (32)$$

On the squarefree physical carrier, the four-corner target sign collapses to

$$\mu(d_m S_m) \mu(d_n S_n) = \mu(a) \mu(b) \mu(u) \mu(v). \quad (33)$$

The common contents g and q disappear. More importantly, (33) is constant in the affine parameter z . The prime ray contributes masks, endpoints, native constraints, and complex amplitudes, but no new Möbius oscillation.

Let $\mathcal{K}_{\mathfrak{a},q}(u, v)$ be the complete ray sum for the fixed full anchor $\mathfrak{a}$, with every physical coefficient and native label retained and with absent rays assigned zero. Aggregate the divisor slices before taking absolute values and impose the literal unequal-residual condition by setting

$$\mathcal{K}_{\mathfrak{a}}^{\circ}(u, v) := \mathbf{1}_{(u,v) \neq (b,a)} \sum_{q \mid |H|} \mathcal{K}_{\mathfrak{a},q}(u, v). \quad (34)$$

The literal contribution of that anchor is exactly

$$\mathfrak{X}_{\mathfrak{a}} = \mu(a) \mu(b) \sum_{\substack{u,v \geq 1 \\ (u,v)=1}} \mu(u) \mu(v) \mathcal{K}_{\mathfrak{a}}^{\circ}(u, v). \quad (35)$$

The complete $\text{tr}(ER)$ is obtained only after summing these anchored contributions over the actual rows and full same-residual anchors. The divisor sum should be aggregated before absolute values whenever the literal phases allow it.

6.2 The physical type check

Write

$$\alpha_{\iota, m} = \frac{\beta_{\iota, m}}{\sqrt{W_m}}$$

for the complete signed and phased coefficient of native key ι in row m , including atomic normalization. An off-diagonal erasure entry is the two-corner object

$$E_{mn} = \sum_{\iota_E} \overline{\alpha_{\iota_E, m}} \alpha_{\iota_E, n}, \quad (36)$$

with the unequal-residual restriction understood. By contrast, fixing a same-residual anchor $\mathbf{a} = \iota_R$ gives the four-corner scalar

$$\mathfrak{X}_{\mathbf{a}} = \sum_{\iota_E} \overline{\alpha_{\iota_E, m}} \alpha_{\iota_E, n} \overline{\alpha_{\iota_R, n}} \alpha_{\iota_R, m}, \quad (37)$$

which is the literal quantity parametrized in (35). Hence

$$\mathrm{tr}(ER) = 2\Re \sum_{m < n} \sum_{\mathbf{a} \in \mathcal{R}_{m, n}} \mathfrak{X}_{\mathbf{a}}. \quad (38)$$

In terms of the unnormalized coefficients β , the four-corner amplitude in (37) already contains $1/(W_m W_n)$.

Proposition 6.1 (Four corners are not an erasure entry). *Suppose $|\mathfrak{X}_{\mathbf{a}}| \leq \mathcal{B}_{\mathbf{a}}$ for every literal anchor. Then*

$$|\mathrm{tr}(ER)| \leq 2 \sum_{m < n} \sum_{\mathbf{a} \in \mathcal{R}_{m, n}} \mathcal{B}_{\mathbf{a}}. \quad (39)$$

This premise alone implies no entrywise bound for E_{mn} , no row-star bound, and no Perron or operator-norm bound for E . Dividing $\mathcal{B}_{\mathbf{a}}$ again by $\sqrt{W_m W_n}$ is a second, unjustified normalization.

Proof. Equation (39) follows from (38) and the triangle inequality. The remaining claims follow already from the abstract example $R = 0$, $E \neq 0$: every four-corner ER target vanishes, while E can have arbitrarily large norm. Thus no inequality involving only these targets can dominate the entries or norm of E . $\square$

Proposition 6.2 (No ray-sign door). *No cancellation based only on changes of the Möbius sign along z can bound (35).*

Proof. For fixed a, b, q, u, v , the sign in (33) is independent of z . Any cancellation along the ray must therefore come from the literal complex amplitude, not from a changing Möbius factor. $\square$

The divisor, determinant, and affine-ray algebra is L0. Its exact reindexing of the full native rectangle is L1. No cancellation is asserted by TPC-70 [10].

6.3 TPC-71: abstract diagonal lift and two-dimensional Abel summation

The primitive sign has the elementary but decisive identity

$$\mathbf{1}_{(u, v)=1} \mu(u) \mu(v) = \mu(uv). \quad (40)$$

For a finitely supported kernel K , Möbius inversion of the coprimality condition gives the exact diagonal divisor lift

$$\sum_{\substack{u, v \geq 1 \\ (u, v)=1}} \mu(u) \mu(v) K(u, v) = \sum_{d \geq 1} \mu(d) \sum_{\substack{a, b \geq 1 \\ (a, d)=(b, d)=1}} \mu(a) \mu(b) K(da, db). \quad (41)$$

Define the restricted Mertens function

$$M_d(x) := \sum_{\substack{n \leq x \\ (n, d)=1}} \mu(n). \quad (42)$$

Extend $K^{[d]}(a, b) = K(da, db)$ by zero outside its finite support, and put

$$\mathcal{V}_\square(K^{[d]}) := \sum_{a, b \geq 1} |\Delta_1 \Delta_2 K^{[d]}(a, b)|. \quad (43)$$

Two-dimensional Abel summation then gives the exact fixed- d identity

$$\sum_{\substack{a, b \geq 1 \\ (a, d) = (b, d) = 1}} \mu(a) \mu(b) K(da, db) = \sum_{x, y \geq 1} M_d(x) M_d(y) \Delta_1 \Delta_2 K^{[d]}(x, y). \quad (44)$$

Zero extension places the edge and corner terms inside the mixed difference. Omitting the boundary jumps would give a false certificate.

For

$$\mathfrak{M}_d(X) = \max_{1 \leq x \leq X} |M_d(x)|,$$

with $\mathfrak{M}_d(X) = 0$ when $X < 1$, define the exact diagonal tail

$$\mathcal{T}_D(K) := \sum_{d > D} |\mu(d)| \sum_{x, y \geq 1} |\mu(dx) \mu(dy) K(dx, dy)|. \quad (45)$$

Truncation at $d \leq D$ yields

$$\left| \sum_{\substack{u \leq U, v \leq V \\ (u, v) = 1}} \mu(u) \mu(v) K(u, v) \right| \leq \sum_{d \leq D} \mathfrak{M}_d(U/d) \mathfrak{M}_d(V/d) \mathcal{V}_\square(K^{[d]}) + \mathcal{T}_D(K). \quad (46)$$

This tail retains the actual support and coefficients. The ambient specialization

$$\mathcal{T}_D(K) \leq \|K\|_\infty \sum_{d > D} \left\lfloor \frac{U}{d} \right\rfloor \left\lfloor \frac{V}{d} \right\rfloor \leq \|K\|_\infty \frac{UV}{D} \quad (47)$$

is only an optional upper bound and may be much larger than the exact tail.

For every fixed d , the prime number theorem gives

$$M_d(x) = o_d(x). \quad (48)$$

Consequently, a bounded family satisfying

$$\mathcal{V}_\square(K^{[d]}) \ll_d \|K\|_\infty$$

for every fixed d has

$$\sum_{\substack{u \leq U, v \leq V \\ (u, v) = 1}} \mu(u) \mu(v) K(u, v) = o(\|K\|_\infty UV) \quad (49)$$

as $\min(U, V) \rightarrow \infty$. In particular, the unweighted primitive rectangle cancels unconditionally. This use of prime number theory is standard; see, for example, Iwaniec and Kowalski [1].

6.4 Why the smooth theorem is not the physical theorem

The physical denominator is not the ambient area $\|K\|_\infty UV$. It is an actual anchored, trace-sector, or atomic mass, and the kernel can contain hard prime-line masks. Therefore the next physical requirements are:

- (V1) a mass-normalized bound for $\mathcal{V}_\square((\mathcal{K}_a^\circ)^{[d]})$, after the literal q -aggregation in (34);
- (V2) the exact tail $\mathcal{T}_D(\mathcal{K}_a^\circ)$ small relative to the same physical mass;
- (V3) justified uniformity in the divisor cutoff and over anchors, blocks, rows, and complete keys;

- (V4) summation over every literal anchor and row pair to control the complete $\text{tr}(ER)$, without losing an unaffordable multiplicity; and
- (V5) for an operator theorem, either a new direct representation of the two-corner entries (36), row-local anchored sums that dominate the relevant stars, or a separate expansion controlling complete decorated moments.

Sign-selected hard masks show that bounded amplitude alone cannot imply these conditions.

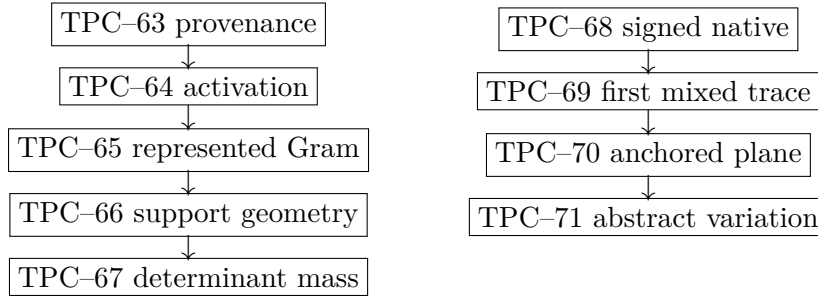
Theorem 6.3 (Primitive-frontier classification). *TPC-70 and TPC-71 close the algebraic reduction from a cross-key rectangle to (46) and prove cancellation for regular kernels. They do not prove a fixed-power or mass-normalized saving for the actual physical kernel. Their physical target is the anchored mixed scalar $\mathfrak{X}_a$, so the reduction does not produce an erasure-entry, star, Perron, or operator certificate.*

Proof. The exact reduction is (35)–(46). The regular-kernel corollary normalizes by $\|K\|_\infty UV$. No audited theorem bounds the actual kernel’s mixed variation or tail relative to the physical mass, nor sums all literal anchors and rows at an endpoint-quality scale. Moreover, proposition 6.1 rules out promotion of that scalar bound to an entrywise operator majorant without a new representation. The missing implications are therefore new quantitative and type-correct premises, not consequences of the regular-kernel theorem. $\square$

7 The dependency graph

7.1 Two branches and one merge

The audited chain is



The left branch seeks to control the represented map A . The right column starts from the missing map M , but after TPC-68 it descends to the scalar sector $\text{tr}(ER)$; it does not return to an operator bound for M . The full maps merge through

$$B = A + M$$

and a shorted comparison relative to A .

7.2 The exact reassembly object

Let

$$N_A = \text{Ker } A, \quad W_{\text{nuis}} = B(N_A) = M(N_A).$$

The direct represented-normalized Gram is not the final uniform comparison, because vectors in N_A can produce physical output through M . The exact shorted Gram has the form

$$G_{\text{sh}} = G_{\text{dir}} - C_N, \quad C_N \succeq 0, \tag{50}$$

following the classical shorted-operator framework [2] and the TPC-68 specialization [8].

The native Gram $I + R + E$ controls an atomic-normalized part of M^*M . It is not automatically G_{dir} , which is normalized by represented energy, and it does not control C_N .

Theorem 7.1 (Dependency theorem). *A uniform endpoint certificate from the audited architecture requires, in one literal carrier and one quantifier scope:*

- (i) a positive terminal activation arrow;
- (ii) a represented lower frame for A ;
- (iii) a missing-mass comparison;
- (iv) an upper bound for the complete missing-native Gram;
- (v) a negligible or explicitly charged nuisance penalty C_N ;
- (vi) boundary and tail closure; and
- (vii) a total physical loss strictly below $1/400$.

No proper subset of these requirements is supplied as the full endpoint theorem by TPC-63–TPC-71.

Proof. TPC-64 shows that a terminal-inactive row kills the uniform first arrow. TPC-65 identifies the represented lower constant with the actual activated Gram and separates label erasure. TPC-67 gives only a conditional determinant route to the second arrow. TPC-68 provides exact abstract certificates for the fourth arrow but no physical arithmetic premise. TPC-69–TPC-71 analyze only the first mixed trace, and [proposition 6.1](#) prevents that scalar analysis from being promoted to the fourth arrow. The mass-degree comparison in TPC-68 requires the third arrow, while (50) shows that the fifth is independent. None of these papers proves the downstream boundary and tail estimates. Finally, TPC-63–TPC-65 record the strict endpoint budget. Each item is therefore a distinct necessary input in the declared staged architecture. $\square$

7.3 Non-substitution laws

Corollary 7.2 (No cross-branch substitution). *The following implications are invalid without additional hypotheses:*

$$\begin{aligned}
& \text{determinant mass} \not\Rightarrow \text{missing-native operator bound}, \\
& \text{primitive-plane scalar cancellation} \not\Rightarrow \text{represented lower frame}, \\
& \text{anchored primitive-plane bound} \not\Rightarrow \text{entrywise erasure bound}, \\
& \|R + E\| = o(1) \not\Rightarrow \|C_N\| = o(1), \\
& \text{tr}(ER) = o(1) \not\Rightarrow \text{uniform row-star saving}.
\end{aligned}$$

Proof. The first two statements concern different maps. The third is [proposition 6.1](#). The fourth follows because C_N depends on $B(N_A)$, which is not determined by the canonical representative of M on $(\text{Ker } A)^\perp$. The fifth is disproved by orthogonal direct sums of reinforcing and cancelling local stars, as in the TPC-69 countermodels [9]. $\square$

8 The strict endpoint ledger

8.1 One global accounting

The complete inherited fixed-power loss is

$$\boxed{\Lambda_{\text{tot}} := \lambda_{\text{sel}} + \lambda_{\text{occ}} + \lambda_{\text{blk}} + \lambda_{\text{term}}^{\text{eff}} + \lambda_{\text{sing}} + \lambda_H + \lambda_{\text{aff}} + \lambda_{\text{reasm}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}}.} \tag{51}$$

Every symbol in (51) denotes the nonnegative optimal fixed-power loss of one distinct physical comparison in the declared scope; it is $+\infty$ when a mandatory comparison constant vanishes infinitely often. The dictionary is:

- λ_{sel} canonical selection from the parent input energy;
- λ_{occ} retention of the required occupied or activated support;

λ_{blk} localization to the actual compatible block or component;
 $\lambda_{\text{term}}^{\text{eff}}$ true terminal activation;
 λ_{sing} the weighted singleton-occupancy comparison;
 λ_H native residual-label erasure on the represented image;
 λ_{aff} base-represented-high affine transversality;
 λ_{reasm} physical label erasure and shorted reassembly;
 $\lambda_{\text{bdry}}, \lambda_{\text{tail}}$ final boundary restriction and tail reconstruction.

The endpoint gate is

$$\Lambda_{\text{tot}} < \frac{1}{400}. \quad (52)$$

Equality is not accepted by the exponent-only route.

The terminal loss satisfies

$$\lambda_{\text{term}}^{\text{eff}} \leq \min\{\lambda_{\text{term}}^{\text{lit}} + \lambda_{\text{infl}}, \lambda_{\text{cap}}\}, \quad (53)$$

when the two TPC-64 certificates are available. No equality is asserted; a direct terminal theorem can be better. The two certificates in the minimum must hold in the same declared coefficient and quantifier scope.

8.2 Coarse and fine notations

TPC-65 compresses part of the chain to

$$\Gamma_X \geq \tau_X \beta_X \delta_X, \quad \Gamma_X \geq X^{-(\lambda_{\text{act}} + \lambda_{\text{rep}} + \lambda_{\text{reasm}}) + o(1)}. \quad (54)$$

This is not an additional loss on top of (51). It is a coarser grouping of some of its arrows. Here λ_{act} is the aggregate of the distinct selection, occupancy, block, terminal, and singleton arrows actually used on the activation side; λ_{rep} is the represented lower comparison; and λ_{reasm} is the physical shorted-reassembly arrow. The represented comparison equals or is bounded using $\lambda_H + \lambda_{\text{aff}}$ only when those are the same physical arrows in the same scope. In particular:

- parent inflation belongs to activation;
- a represented exponent belongs to the represented arrow once;
- physical label erasure and missing-native perturbation belong to reassembly;
- exact coordinate relabeling costs zero only when it is unitary or has $X^{o(1)}$ condition number.

Two inherited no-double-charge rules are especially easy to violate. If selection, occupancy, or block losses were already used inside a direct terminal ratio $T/\tilde{P}$, they are removed from the pre-terminal sum. If the represented comparison λ_{rep} merely repackages the physical arrow already encoded by $\lambda_H + \lambda_{\text{aff}}$, it is not charged again through λ_{reasm} .

8.3 Mass-degree feasibility is not a second ledger

Let A_X and M_X be the represented and missing-native maps on the same raw physical coefficient space $\mathcal{K}_X$. With missing atoms $\mathcal{W}_M$, define

$$W_M(m) := \sum_{\substack{w \in \mathcal{W}_M \\ m(w)=m}} |\beta_w|^2, \quad D_M := \text{diag}(W_M(m)), \quad S_M := \text{Ran } D_M. \quad (55)$$

Thus D_M uses raw physical atomic mass. The missing-native synthesis M_X is built from the same atoms, so, with P_{S_M} the orthogonal projection onto S_M ,

$$M_X(I - P_{S_M}) = 0. \quad (56)$$

If $S_M \neq \{0\}$, set

$$\kappa_M := \sup_{\substack{z \in \mathcal{K}_X \\ \langle D_M z, z \rangle > 0}} \frac{\|M_X z\|^2}{\langle D_M z, z \rangle} = \|M_X D_M^{\dagger/2}|_{S_M}\|^2; \quad (57)$$

set $\kappa_M = 0$ when the missing carrier is empty. In particular,

$$\|M_X z\|^2 \leq \kappa_M \langle D_M z, z \rangle. \quad (58)$$

This is the restricted physical Rayleigh constant, not an ambient degree, $\|M_X\|^2$, or a quotient for a vector selected after the estimate.

Let $\mathcal{D}_{\text{ref}}$ be one nonnegative reference form. Fix either the uniform represented scope $z \in (\ker A_X)^\perp$ or one predeclared distinguished vector. In the distinguished case one may use the corresponding fixed-vector Rayleigh quotient, but it must not be reported as the uniform κ_M . Suppose, throughout the chosen scope,

$$\kappa_M \leq X^{\nu_M + o(1)}, \quad (59)$$

$$\langle D_M z, z \rangle \leq X^{-\sigma_M + o(1)} \mathcal{D}_{\text{ref}}(z), \quad (60)$$

$$\|A_X z\|^2 \geq X^{-\lambda_{\text{rep}} + o(1)} \mathcal{D}_{\text{ref}}(z). \quad (61)$$

Then

$$\frac{\|M_X z\|^2}{\|A_X z\|^2} \leq X^{\nu_M + \lambda_{\text{rep}} - \sigma_M + o(1)} \quad (62)$$

whenever $A_X z \neq 0$. Thus

$$\boxed{\sigma_M > \nu_M + \lambda_{\text{rep}}} \quad (63)$$

makes the missing perturbation $o(1)$.

When (59)–(61) are used only to prove (63), their exponents are diagnostic inputs to λ_{reasm} ; they are not added a second time to (51). If the comparison gives a genuine nonzero loss instead, that resulting physical loss is charged once.

The nuisance condition

$$\|C_N\| = o(1) \quad (64)$$

remains separate.

Proposition 8.1 (No-double-charge law). *Choose one decomposition of the actual physical chain into composable arrows. Algebraic identities and isometric or $X^{o(1)}$ -conditioned relabelings cost zero. The fixed-power loss of each non-isometric physical arrow is entered exactly once. Any refined exponents used only to certify one arrow are not added again.*

Proof. Successive lower comparisons multiply, so their logarithmic losses add. Factoring one arrow internally and then adding both its internal losses and its already-derived external loss counts the same ratio twice. Conversely, omitting a distinct non-isometric arrow removes a necessary comparison. This proves the stated accounting rule. $\square$

8.4 Current endpoint status

The algebraic identities and isometric or subpower-conditioned relabelings in TPC–63–TPC–71 cost zero. Their non-isometric physical comparisons, including parent inflation when used, must be charged once. The block does not assign endpoint-quality values simultaneously to $\lambda_{\text{term}}^{\text{eff}}$, λ_{rep} , ν_M , σ_M , the nuisance term, and the downstream boundary and tail. Therefore (52) has not been verified.

9 Closed interfaces, failed mechanisms, and open doors

The word “closed” has two meanings. An interface may be solved positively, or a proposed route may be ruled out. Mixing these meanings obscures the remaining work.

9.1 Positively closed interfaces

Interface	Exact closure	Level
Provenance	Minimal faithful pair relation when it exists; literal hypergraph fallback when it does not; three separate faithfulness tests.	L0/L1
Activation calculus	Exact P_m, H_m, T_m , parent inflation, hybrid repair, and sharp capacity certificate.	L0/L1
Represented range	Exact activated row Gram, repair invariance, kernel fusion, and label-erasure boundary.	L0/L1
Pair support	Unit diagonal and critical fixed-row-pair reduced codegree $X^{o(1)}$.	L1
Determinant conversion	Peeling, cycle activity, Cauchy–Binet mass, and the componentwise determinant-to-gap inequality.	L0/L1
Native operator	Exact pair, star, Perron, moment, and shorted-Gram certificates on the complete missing carrier.	L0/L1
Mixed sectors	Key-local vanishing, cyclic erasure-count formula, odd residual boundary, and nonnegative quadratic erasure response.	L0/L1
Primitive plane	Exact divisor and affine-ray parametrization of the first cross-key coefficient.	L0/L1
Variation certificate	Exact diagonal divisor lift, two-dimensional Abel formula, tail bound, and smooth-kernel cancellation.	L0/L1

9.2 Mechanisms closed by obstruction

- (F1) **Inherited single band.** The old freezing certificate misses the critical face, and its retuning already exceeds $1/400$.
- (F2) **Label-count dilution.** $O(\log X)$ labels give bookkeeping economy, not a lower bound after coherent erasure.
- (F3) **Capacity transport.** Row capacity does not imply spectral capacity without a reducing or basis-transport theorem.
- (F4) **Support-only Gram closure.** Low pair codegree does not control γ -multiplicity, energy diffuseness, or cycle holonomy.
- (F5) **Same-residual randomness.** Its target sign is a removable row gauge on closed walks.
- (F6) **One-key mixed cancellation.** $\text{tr}(E_i R_i^a) = 0$ identically.
- (F7) **Negative quadratic correction.** The full $S_{2q,2}$ is nonnegative.
- (F8) **Prime-ray Möbius oscillation.** The primitive sign is constant along the affine ray parameter.
- (F9) **Bounded-amplitude variation.** Hard sign-selected masks show that $\|K\|_\infty$ alone does not control the primitive-plane sum.

(F10) **Four-corner-to-entry promotion.** An anchored primitive-plane target contributes to $\text{tr}(ER)$; it is not a matrix entry of E , and cannot be inserted directly into a pair, star, or Perron operator certificate.

Each failure is local to the named mechanism. None is evidence that the twin-prime conjecture is false.

9.3 Open doors

Door	Required theorem	Current status
Terminal activation	(5) or the exact distinguished capacity target in the same downstream scope.	Open L2
Represented frame	Physical unlabelled Gram lower bound, for example an endpoint-quality worst-component determinant mass.	Open L2
Primitive variation	Mass-normalized mixed variation and diagonal tail for the actual aggregated kernel $\mathcal{K}_a^\circ(u, v)$.	Open L2
Anchored mixed trace	Mass-normalized variation and tail, followed by the complete sum over divisors, anchors, row pairs, keys, and blocks in (39).	Open L2
Operator carrier	A direct two-corner representation of E_{mn} , row-local anchored bounds dominating the relevant stars, or complete decorated-moment estimates for the full missing-native operator.	Open L2
Missing mass	(60) in the same reference form as the represented lower bound.	Open L2
Nuisance shorting	$\ C_N\ = o(1)$, or an explicitly charged nonzero loss.	Open L2
Boundary and tail	Reconnection of the certified physical packet to the endpoint target, with all remaining losses recorded.	Open L2

Theorem 9.1 (No completed arithmetic column). *No theorem among TPC-63–TPC-71 proves one of the open doors above with all six physical invariants of section 2 and simultaneously closes the endpoint chain.*

Proof. TPC-63–TPC-65 explicitly leave activation and unlabelled representation open. TPC-66 proves support codegree but not a Gram gap. TPC-67 leaves actual determinant mass open. TPC-68–TPC-69 state their pair, star, walk, missing-mass, and nuisance premises conditionally. TPC-70 gives an exact reindexing without cancellation. TPC-71 proves regular-kernel cancellation but explicitly leaves physical mass-normalized variation, tail, and complete anchor summation open. Its four-corner scalar target cannot be used as an entrywise operator majorant by proposition 6.1. Thus no audited theorem satisfies the stated conjunction. $\square$

10 Kill criteria and their logical strength

10.1 Object and quantifier stops

- (K1) **Wrong object.** Removing side, external block, opposite row, γ , physical time, or another still-orthogonal native label estimates a different operator.
- (K2) **Wrong support.** Replacing actual active support by a terminal image, reduced cell, or coefficient-dependent subset changes the carrier.

- (K3) **Wrong norm.** Fiber-normalized atoms cannot replace raw atomic normalization.
- (K4) **Wrong quantifier.** A fixed row pair, density-one family, averaged shift, smoothed family, or partition chosen after the coefficient vector is known does not prove the uniform fixed- h_0 statement.
- (K5) **Wrong denominator.** A lower bound relative to ambient area, a repaired parent, or a labelled norm cannot be inserted relative to physical mass without an exact comparison.

These failures invalidate the proposed proof or promotion. They are not evidence that the underlying physical quantity is large or that every route to it fails.

10.2 Hard structural stops

- (H1) If a required row has $T_m = 0 < \tilde{P}_m$, the uniform terminal activation route is dead on any class containing that row.
- (H2) If the represented map restricted to the declared activated domain or quotient has a kernel, its uniform lower-frame constant there is zero. A kernel outside that domain is irrelevant, and a distinguished vector remains a different question.
- (H3) If a full-output collision component has genuine rank deficiency, its determinant mass is zero.
- (H4) If the actual shorted Gram has a kernel, uniform reassembly through the declared architecture is impossible.
- (H5) If the complete physical loss is at least $1/400$, including equality, the present exponent-only endpoint implication stops. A different constant-sensitive theorem would be a new argument.

10.3 Certificate-local stops

The next failures are weaker.

- (C1) A pair majorant with $\tau < \delta$ fails to produce a subpower pair/Perron bound by that certificate. The positive exponent $\delta - \tau$ may still be absorbed by (63), and a later star or moment route remains open.
- (C2) A star saving with $\sigma < \delta/2$ fails to produce a subpower direct-star bound. The exponent $\delta/2 - \sigma$ may still be compatible with the mass-degree gate, and complete walks remain open.
- (C3) Failure at one moment length does not kill other lengths; a fixed- q estimate with uncontrolled constants does not prove a growing- q theorem.
- (C4) Matching activity at least one kills the phase-uniform dominant matching certificate, not necessarily the minor.
- (C5) A superpolynomially small product over many orthogonal components does not kill the represented gap. The relevant obstruction is the actual worst component.
- (C6) Large mixed variation kills the TPC-71 Abel certificate at the required scale, not necessarily the literal signed sum.
- (C7) A large diagonal tail for one cutoff D kills that truncation choice; it does not rule out a different exact decomposition.
- (C8) If $\sigma_M \leq \nu_M + \lambda_{\text{rep}}$, the phase-blind mass-degree certificate does not prove a small perturbation in that scope. It does not prove that the actual missing perturbation is large.
- (C9) Infinite parent inflation or new-only parent rows kill the literal-to-repaired inflation transfer. A direct capacity or literal-hypergraph activation route can remain; the hard uniform stop is $T_m = 0 < \tilde{P}_m$ on a required row.

(C10) Small global $\text{tr}(ER)$ does not certify any uniform local star.

(C11) Small anchored four-corner targets do not bound the two-corner entries of E ; using them as such is a wrong-object failure, not an insufficient exponent.

10.4 Primitive-plane route stops

The current narrow route has five specific tests.

- (P1) The complete divisor and native-key census must have only controlled multiplicity. Compressing keys before the census is forbidden.
- (P2) The zero-extended mixed variation must include every interior, edge, and corner jump. Omitting boundary jumps invalidates the Abel identity.
- (P3) The exact tail $\mathcal{T}_D(K)$ must be small relative to actual mass. Failure of the ambient specialization $\|K\|_\infty UV/D$ kills only that ambient majorant, not the exact support tail or another decomposition.
- (P4) The anchor bounds must survive the complete sum in (39). Even if they do, the output is a mixed-trace theorem, not an operator theorem. Failure after anchorwise triangle summation kills that aggregation; it does not exclude cancellation across anchors or a complete-moment route.
- (P5) The four-corner normalization $1/(W_m W_n)$ must be retained literally and charged once. A further division by $\sqrt{W_m W_n}$ changes the scale.

Tests P1, P2, and P5 validate the object and identity; violating them changes the theorem rather than proving a negative result. P3 and P4 are certificate-local quantitative gates.

Proposition 10.1 (Meaning of a route obstruction). *A sharp obstruction or non-implication theorem for the actual declared certificate is a publishable negative theorem about that proof mechanism. Mere failure to prove a sufficient estimate is an unresolved gate. Neither statement is a parity theorem or has any implication for the truth of a prime-pair conjecture.*

Proof. Each certificate uses a sufficient inequality obtained by discarding some literal information. Failure to establish it does not imply failure of the underlying quantity. A genuine negative result requires a lower obstruction, an exact type mismatch, or an extremal theorem proving necessity. The audited papers supply countermodels separating these implications. $\square$

11 Revised next-stage routes

11.1 Route A: anchored primitive-plane variation

For one row pair and one full same-residual anchor $\mathbf{a}$, let $U_{\mathbf{a}}, V_{\mathbf{a}}$ be the actual primitive-coordinate extents of the aggregated literal kernel $\mathcal{K}_{\mathbf{a}}^\circ$ from (34). Define the exact Abel cost

$$\mathfrak{A}_{\mathbf{a}}(D) := \sum_{d \leq D} \mathfrak{M}_d(U_{\mathbf{a}}/d) \mathfrak{M}_d(V_{\mathbf{a}}/d) \mathcal{V}_{\square}((\mathcal{K}_{\mathbf{a}}^\circ)^{[d]}) + \mathcal{T}_D(\mathcal{K}_{\mathbf{a}}^\circ). \quad (65)$$

Define the anchored plane mass, with absolute values taken after the exact q - and ray-aggregation, by

$$\mathfrak{W}_{\mathbf{a}} := \sum_{u, v \geq 1} |\mathcal{K}_{\mathbf{a}}^\circ(u, v)|. \quad (66)$$

Because $\mathcal{K}_{\mathbf{a}}^\circ$ already contains the normalized four-corner amplitudes,

$$|\mathfrak{X}_{\mathbf{a}}| \leq \mathfrak{W}_{\mathbf{a}}.$$

This mass can be smaller than the sum of absolute pre-aggregation ray atoms, and it is not automatically the global trace or raw atomic mass. A first useful arithmetic target is

$$\mathfrak{A}_a(D_X) \leq X^{-\tau_{\text{plane}} + o(1)} \mathfrak{W}_a \quad (67)$$

uniformly over every active anchor, pair, block, and key in a declared scope. The physical conclusion comes only after the complete anchor and row sum in (39). If D_X grows, (67) additionally requires a proved restricted-Mertens estimate uniform throughout $d \leq D_X$. With only fixed- d input, the safe formulation fixes D , takes the physical scale limit, and only then lets $D \rightarrow \infty$.

This target must be attacked in the following order:

- (A1) compute the actual support and form the literal q -aggregate $\mathcal{K}_a^\circ$ before absolute values;
- (A2) compute its zero-extended mixed variation, including bulk, edge, and corner jumps;
- (A3) estimate the exact tail $\mathcal{T}_D(\mathcal{K}_a^\circ)$; use (47) only when that specialization is sharp enough;
- (A4) choose a fixed cutoff or justify a growing D_X against the physical mass rather than ambient area;
- (A5) test whether the resulting saving survives complete trace aggregation.

If a sharp lower obstruction proves that the exact mass-normalized cost in (65) is necessarily trivial on a nonnegligible anchor family, this Abel certificate is closed for that carrier. Failure of the ambient upper bound alone closes only the ambient-tail specialization, and failure to prove the sufficient inequality is an unresolved gate rather than a negative theorem.

11.2 Route B: build a genuine operator carrier

Equation (67) does not give an entrywise pair bound. To invoke the TPC-68 pair/star machinery, one must first construct a direct primitive or other arithmetic representation of the two-corner entry (36). If that new representation gives an entrywise saving $X^{-\tau_{\text{ent}} + o(1)}$ and the actual pair graph has degree exponent δ , the TPC-68 pair route requires

$$\tau_{\text{ent}} \geq \delta \quad \text{for a subpower norm,} \quad \tau_{\text{ent}} > \delta \quad \text{for decay.} \quad (68)$$

If square summation improves the local estimate to star exponent σ , the threshold becomes

$$\sigma \geq \delta/2 \quad \text{or} \quad \sigma > \delta/2. \quad (69)$$

Below these thresholds the corresponding certificate supplies neither a subpower bound nor decay by itself. It does not kill the whole operator route: a positive norm exponent may still be absorbed by the mass-degree gate (63).

A second bridge may instead prove row-local anchored bounds that dominate the actual stars without first identifying every entry. This requires a new row-local crosswalk and cannot be inferred from the global trace. A third bridge is to estimate complete decorated moments with constants uniform in the growing length. The moment option requires a new crosswalk for every decorated sector actually used; the first mixed trace cannot stand in for the full moment.

The global trace $\text{tr}(ER)$ is useful as a diagnostic of the linear sector, but it is not an operator aggregation target. The next paper must therefore choose explicitly among a direct two-corner entry route, a row-local anchored-star bridge, and a full-moment route. A fixed-row star or reducible-safe Perron majorant becomes available only after a literal carrier dominating that object has been constructed.

11.3 Route C: represented-frame closure in parallel

The primitive-plane branch should not monopolize the program. Independently, one needs:

- (R1) a terminal-candidate lower theorem giving (5) or its distinguished analogue;

- (R2) a physical represented Gram lower bound, through determinant mass, a comparison matrix, or a different exact certificate;
- (R3) an explicit physical label-erasure or fusion-angle estimate.

The determinant route is alive precisely because its conversion is exact, but it needs actual matching, pivot, and cycle-activity input. If the worst actual component determinant mass is proved superpolynomially small, that closes this certificate at endpoint scale while leaving other represented certificates open.

11.4 Route D: final mass-degree and shorting merge

Only after Routes A–C give compatible inputs should one test

$$\sigma_M > \nu_M + \lambda_{\text{rep}} \quad (70)$$

and

$$\|C_N\| = o(1). \quad (71)$$

Both must use the same reference energy, carrier, support, coefficient scope, and fixed shift. The endpoint ledger is then evaluated once, with boundary and tail attached last.

11.5 Fallbacks if the anchored trace door closes

If the physical primitive-plane variation is too large, or if its residual boundary is frozen on the actual support, the remaining signed alternatives are:

- (i) higher one-erasure polygons;
- (ii) cancellation from literal complex phases;
- (iii) a direct shorted reassembly estimate; or
- (iv) a coefficient-specific distinguished theorem with its scope stated explicitly.

The complete quadratic sector is not a fallback source of negative compensation by (25).

11.6 Stage schedule

A disciplined next block can be organized as follows.

Stage 1. Anchored physical variation census and boundary decomposition.

Stage 2. A uniform or obstructed first-mixed-trace estimate.

Stage 3. A direct two-corner, row-local anchored-star, or complete-moment operator crosswalk.

Stage 4. Pair-to-star/Perron aggregation or uniform moment control, according to the carrier established in Stage 3.

Stage 5. Parallel terminal activation and represented determinant work.

Stage 6. Missing-mass and nuisance-shortening comparison.

Stage 7. Boundary, tail, and strict endpoint audit.

At every stage, success means either a literal exponent improvement, a necessary interface closure, or an exact actual-carrier obstruction. A model refinement without one of these outputs should not be reported as movement toward the parity endpoint.

12 Claim ledger and status line

12.1 Exact conclusions of this audit

This paper proves the following statements about the audited architecture.

- (1) TPC-63–TPC-71 form two logically distinct branches, represented and missing-native, which meet only at physical reassembly.
- (2) Every completed theorem in the block is classified as L0, L1, or an explicitly conditional L2 premise.
- (3) The inherited single-band certificate, same-residual sign route, one-key mixed route, negative quadratic correction, and ray-sign route are closed by exact obstructions.
- (4) The first surviving signed coefficient is the primitive-plane sum (35).
- (5) TPC-71 gives the exact certificate (46) and unconditional cancellation for regular kernels, but not for the actual mass-normalized physical kernel.
- (6) The physical primitive-plane target is the anchored four-corner scalar $\mathfrak{X}_a$, and the valid aggregate conclusion is (39). It is not an entrywise erasure majorant.
- (7) The complete endpoint requires every item of theorem 7.1; no audited theorem supplies the full conjunction.
- (8) The common no-double-charge ledger is (51), with strict gate (52).

12.2 Statements not proved

This paper does not prove:

- a positive uniform terminal activation exponent;
- a polynomial lower frame for the physical unlabelled represented map;
- endpoint-quality determinant mass on the worst actual component;
- a mass-normalized mixed-variation bound for the physical primitive-plane kernel;
- a complete endpoint-quality bound for $\text{tr}(ER)$;
- a direct two-corner pair/star/Perron majorant, a row-local anchored-star bridge, or a complete growing-moment saving;
- $\|R + E\| = X^{o(1)}$, much less $o(1)$;
- the missing-mass estimate (60);
- the nuisance estimate $\|C_N\| = o(1)$;
- the strict endpoint inequality for one completed physical chain;
- any estimate using a property special to $h_0 = 2$; or
- a parity-breaking, prime-pair, or twin-prime theorem.

12.3 Ten-paper result table

Paper	Exact advance	Honest status
TPC-63	Canonical provenance and single-band support-certificate no-go.	L0/L1; no L2
TPC-64	True activation, parent inflation, and sharp capacity law.	L0/L1; arithmetic activation open
TPC-65	Activated-range represented Gram and compatible three-stage product.	L0/L1; physical gap open
TPC-66	Unit diagonal and critical reduced pair codegree $X^{o(1)}$.	Genuine L1 support theorem; no operator saving
TPC-67	Componentwise determinant mass converts to represented gap.	L0/L1; actual mass lower bound open
TPC-68	Signed native pair/star/walk certificates and exact shorting.	L0/L1; all arithmetic premises open
TPC-69	Cross-key mixed holonomy, odd residual boundary, quadratic positivity.	L0/L1; linear saving open
TPC-70	Primitive cofactor-plane and frozen affine prime rays.	L0/L1; no cancellation
TPC-71	Divisor lift, two-dimensional Abel certificate, regular-kernel cancellation, and anchored mixed-trace crosswalk.	L0/L1; direct operator promotion unavailable
TPC-72	Dependency theorem, four-corner type boundary and non-substitution theorem, unified ledger, revised routes.	Audit closure; no new arithmetic saving

12.4 Status line

The ten-paper block moves the program from an ambiguous cofactor-support picture to a falsifiable primitive-plane frontier. It has closed necessary interfaces and eliminated several false sources of cancellation. As of TPC-72, however, the L2 arithmetic column required by the endpoint remains empty. The next advance must control the literal physical primitive-plane variation and tail strongly enough to bound the full anchored mixed trace. A separate advance must then construct a genuine two-corner, row-local anchored-star, or complete-moment operator carrier, while terminal activation and the represented lower frame are advanced in parallel. This is progress in proof architecture, not evidence that the parity barrier has been broken.

References

- [1] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 2004. doi: 10.1090/coll/053.
- [2] Jr. W. N. Anderson and George E. Trapp. Shorted operators. II. *SIAM Journal on Applied Mathematics*, 28(1):60–71, 1975. doi: 10.1137/0128007.
- [3] Liang Wang. Canonical cofactor provenance at a fixed shift: Minimal faithful relations, exact exposure factorization, and the single-band completion barrier, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-63-canonical-cofactor-provenance. Preprint.
- [4] Liang Wang. True terminal activation on canonical multiscale incidences: Pair-lift minimality, parent inflation, and sharp capacity laws, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-64-true-terminal-activation. Preprint.

- [5] Liang Wang. Activated-range represented grams at a fixed shift: Repair invariance, multiblock kernel fusion, and the failure of row capacity transfer, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-65-activated-range-represented-gram. Preprint.
- [6] Liang Wang. Actual exposure and low pair codegree in an activated fixed-shift residual gram: Row singletons, forest-exact lower frames, and cycle-holonomy obstructions, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-66-low-codegree-actual-support. Preprint.
- [7] Liang Wang. Determinant mass on the actual collision core at a fixed shift: Forced matchings, alternating holonomy, and cauchy–binet lower frames, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-67-actual-collision-core-determinant-mass. Preprint.
- [8] Liang Wang. Signed native operator certificates at a fixed shift: Pairing defects, star norms, operator walks, and the shorted reassembly boundary, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-68-signed-native-operator-certificates. Preprint.
- [9] Liang Wang. Erasure-boundary expansions in a fixed-shift native gram: Mixed words, residual parity, and cross-key holonomy, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-69-erasure-boundary-mixed-words. Preprint.
- [10] Liang Wang. Primitive cofactor coordinates for two-key erasure rectangles: Divisor compression, frozen prime rays, and the residual möbius plane, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-70-primitive-cofactor-plane. Preprint.
- [11] Liang Wang. Divisor-lifted möbius certificates on the primitive cofactor plane: Two-dimensional abel summation, coprimality tails, and the physical variation gate, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-71-divisor-lifted-plane-certificates. Preprint.